Phyton For Data Science

Cheat-Sheet Phyton Basic

BecomingHuman.AI

DataCamp

Variables and Data Types

Variable Assignment

```
>>> x=5
>>> x
 5
```

Calculations With Variables

```
>>> x+2                    Sum of two variables
 7
>>> x-2                    Subtraction of two variables
 3
>>> x*2                    Multiplication of two variables
10
>>> x**2                   Exponentiation of a variable
 25
>>> x%2                    Remainder of a variable
 1
>>> x/float(2)             Division of a variable
 2.5
```

Calculations With Variables

str()	'5', '3.45', 'True'	Variables to strings
int()	5, 3, 1	Variables to integers
float()	5.0, 1.0	Variables to floats
bool()	True, True, True	Variables to booleans

Asking For Help

```
>>> help(str)
```

Lists

Also see NumPy Arrays

```
>>> a = 'is'
>>> b = 'nice'
>>> my_list = ['my', 'list', a, b]
>>> my_list2 = [[4,5,6,7], [3,4,5,6]]
```

Selecting List Elements

Index starts at 0

```
Subset
>>> my_list[1]              Select item at index 1
>>> my_list[-3]             Select 3rd last item
Slice
>>> my_list[1:3]            Select items at index 1 and 2
>>> my_list[1:]             Select items after index 0
>>> my_list[:3]             Select items before index 3
>>> my_list[:]              Copy my_list
Subset Lists of Lists
>>> my_list2[1][0]          my_list[list][itemOfList]
>>> my_list2[1][:2]
```

List Operations

```
>>> my_list + my_list
['my', 'list', 'is', 'nice', 'my', 'list', 'is', 'nice']
>>> my_list * 2
['my', 'list', 'is', 'nice', 'my', 'list', 'is', 'nice']
>>> my_list2 > 4
True
```

List Methods

```
>>> my_list.index(a)        Get the index of an item
>>> my_list.count(a)        Count an item
>>> my_list.append('!')     Append an item at a time
>>> my_list.remove('!')     Remove an item
>>> del(my_list[0:1])       Remove an item
>>> my_list.reverse()       Reverse the list
>>> my_list.extend('!')     Append an item
>>> my_list.pop(-1)         Remove an item
>>> my_list.insert(0,'!')   Insert an item
>>> my_list.sort()          Sort the list
```

Numpy Arrays

Also see Lists

```
>>> my_list = [1, 2, 3, 4]
>>> my_array = np.array(my_list)
>>> my_2darray =
np.array([[1,2,3],[4,5,6]])
```

Selecting Numpy Array Elements

Index starts at 0

```
Subset
>>> my_array[1]             Select item at index 1
  2
Slice
>>> my_array[0:2]           Select items at index 0 and 1
  array([1, 2])
Subset 2D Numpy arrays
>>> my_2darray[:,0]         my_2darray[rows, columns]
  array([1, 4])
```

Numpy Array Operations

```
>>> my_array > 3
array([False, False, False, True], dtype=bool)
>>> my_array * 2
array([2, 4, 6, 8])
>>> my_array + np.array([5, 6, 7, 8])
array([6, 8, 10, 12])
```

Numpy Array Operations

```
>>> my_array.shape          Get the dimensions of the array
>>> np.append(other_array)  Append items to an array
>>> np.insert(my_array, 1, 5)  Insert items in an array
>>> np.delete(my_array,[1])  Delete items in an array
>>> np.mean(my_array)       Mean of the array
>>> np.median(my_array)     Median of the array
>>> my_array.corrcoef()     Correlation coefficient
>>> np.std(my_array)        Standard deviation
```

Strings

Also see NumPy Arrays

```
>>> my_string = 'thisStringIsAwesome'
>>> my_string
'thisStringIsAwesome'
```

String Operations

```
>>> my_string * 2
'thisStringIsAwesomethisStringIsAwesome'
>>> my_string + 'Innit'
'thisStringIsAwesomeInnit'
>>> 'm' in my_string
True
```

String Operations

Index starts at 0

```
>>> my_string[3]
>>> my_string[4:9]
```

String Methods

```
>>> my_string.upper()       String to uppercase
>>> my_string.lower()       String to lowercase
>>> my_string.count('w')    Count String elements
>>> my_string.replace('e', 'i')  Replace String elements
>>> my_string.strip()       Strip whitespaces
```

Libraries

Import libraries

```
>>> import numpy
>>> import numpy as np
```

Selective import

```
>>> from math import pi
```

Install Python

ANACONDA — Leading open data science platform powered by Pytho

SPYDER — Free IDE that is included with Anaconda

jupyter — Create and share documents with live code, visualizations, text, ...

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